

Hot Water Distribution - CPVC **FlowGuard Gold** CTS (SDR 11)

System	Charlotte FlowGuard Gold CTS CPVC pipe & fittings, SDR 11
Material	Rigid CPVC (ASTM D1784: pipe 24448 / fittings 23447)
Standard	ASTM D2846
Rating	Up to 180 F at 100 psi
Potable	NSF-61 and NSF-14
Sizes	1/2 in to 2 in CTS
Joints	CPVC cement ASTM F493

Notes

Hot water distribution (CPVC); same system as J12.
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SUBMITTAL FOR CHARLOTTE PIPE® FLOWGUARD GOLD® CPVC COPPER TUBE SIZE (CTS) HOT AND COLD DOMESTIC WATER DISTRIBUTION SYSTEM

Date: _____

Job Name: _____

Location: _____

Engineer: _____

Contractor: _____

► Scope:

This specification covers Copper Tube Size (CTS) CPVC manufactured to standard dimensional ratio (SDR) 11 for hot and cold domestic water distribution. This system is intended for pressure applications where the operating temperature will not exceed 180° F at 100 psi.

► Specification:

Pipe shall be manufactured from virgin rigid CPVC (chlorinated polyvinyl chloride) vinyl compounds with a cell class of 24448 as identified in ASTM D 1784. Fittings shall be manufactured from virgin rigid CPVC (chlorinated polyvinyl chloride) vinyl compounds with a cell class of 23447 as identified in ASTM D 1784.

FlowGuard Gold CTS CPVC pipe and fittings shall conform to ASTM D 2846. Pipe and fittings shall be manufactured as a system and be the product of one manufacturer. All pipe and fittings shall be manufactured in the United States. Pipe and fittings shall conform to NSF International Standards 14 and 61.

► Installation:

Installation shall comply with the latest installation instructions published by Charlotte Pipe and Foundry and shall conform to all applicable plumbing, fire, and building code requirements. Buried pipe shall be installed in accordance with ASTM F 1668. Solvent cement joints shall be made using CPVC cement conforming to ASTM F 493. If a primer is required by local plumbing or building codes, then a primer conforming to ASTM F 656 should be used. The system shall be protected from chemical agents, fire-stopping materials, thread sealant, plasticized-vinyl products or other aggressive chemical agents not compatible with CPVC compounds. The system shall be hydrostatically tested after installation. **WARNING!** Never test with or transport/store compressed air or gas in CPVC pipe or fittings. Doing so can result in explosive failures and cause severe injury or death.

► Referenced Standards:

ASTM D 1784: Rigid Vinyl Compounds
 ASTM D 2846: CPVC Plastic Hot and Cold Water Distribution System
 ASTM F 493: Solvent Cements for CPVC Pipe and Fittings
 ASTM F 1668: Procedures for Buried Plastic Pipe
 NSF Standard 14: Plastic Piping Components & Related Materials
 NSF Standard 61: Drinking Water System Components—Health Effects



 Quarter Bend	 Eighth Bend	 Brass Male Adapter
 Street Quarter Bend	 Male Adapter	 Bushing
 Cap	 Tee	 Coupling

Not all fitting patterns shown

**MINIMUM DIMENSIONS FROM CENTER TO END OF SOCKET
(LAYING LENGTH) FOR CPVC 41, SDR 11 FITTINGS**

NOMINAL SIZE (IN.)	"G" MINIMUM IN. (mm)	"J" MINIMUM IN. (mm)	"N" MINIMUM IN. (mm)
1/2	0.382 (9.70)	0.183 (4.65)	0.102 (2.59)
3/4	0.507 (12.88)	0.235 (5.97)	0.102 (2.59)
1	0.633 (16.08)	0.287 (7.29)	0.102 (2.59)
1 1/4	0.758 (19.25)	0.339 (8.61)	0.102 (2.59)
1 1/2	0.884 (22.45)	0.391 (9.93)	0.102 (2.59)
2	1.134 (28.83)	0.495 (12.57)	0.102 (2.59)

PIPE REFERENCE GUIDE

Product	Sizes Available					
	1/2	3/4	1	1 1/4	1 1/2	2
FlowGuard Gold® CPVC CTS SDR 11	•	•	•	•	•	•